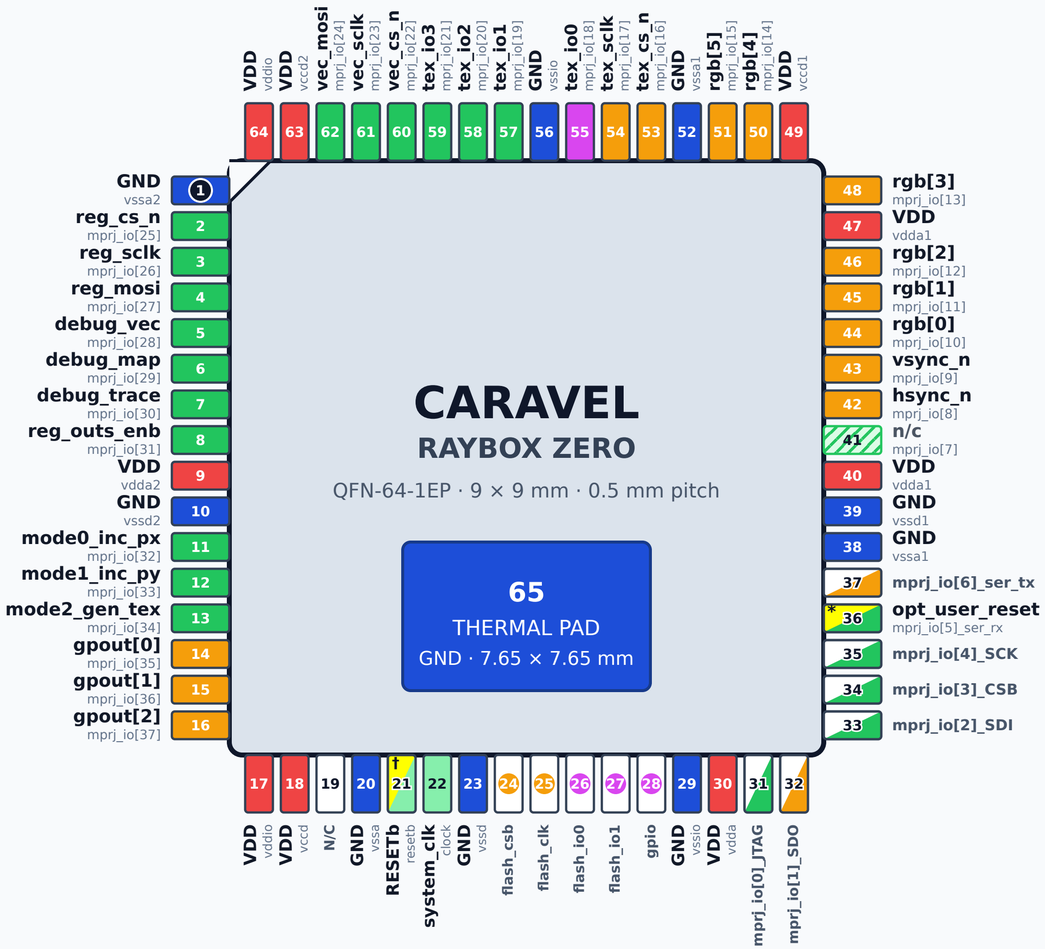
RAYBOX ZERO — CARAVEL QFN64 PINOUT

Top view · GF180 multi-project implementation · mux design ID 0 or 1



Pin	Caravel / U1	Raybox Zero	Dir
1	vssa2	GND	GND
2	mprj_io[25]	reg_cs_n	IN
3	mprj_io[26]	reg_sclk	IN
4	mprj_io[27]	reg_mosi	IN
5	mprj_io[28]	debug_vec	IN
6	mprj_io[29]	debug_map	IN
7	mprj_io[30]	debug_trace	IN
8	mprj_io[31]	reg_outs_enb	IN
9	vdda2	VDD	PWR
10	vssd2	GND	GND
11	mprj_io[32]	mode0_inc_px	IN
12	mprj_io[33]	mode1_inc_py	IN
13	mprj_io[34]	mode2_gen_tex	IN
14	mprj_io[35]	gpout[0]	OUT
15	mprj_io[36]	gpout[1]	OUT
16	mprj_io[37]	gpout[2]	OUT
17	vddio	VDD	PWR
18	vccd	VDD	PWR
19	N/C	—	—
20	vssa	GND	GND
21	resetb	RESETb	IN
22	clock	system_clk	IN
23	vssd	GND	GND
24	flash_csb	—	OUT
25	flash_clk	—	OUT
26	flash_io0	—	I/O
27	flash_io1	—	I/O
28	gpio	—	I/O
29	vssio	GND	GND
30	vdda	VDD	PWR
31	mprj_io[0]_JTAG	—	IN
32	mprj_io[1]_SDO	—	OUT

Pin	Caravel / U1	Raybox Zero	Dir
33	mprj_io[2]_SDI	—	IN
34	mprj_io[3]_CSB	—	IN
35	mprj_io[4]_SCK	—	IN
36	mprj_io[5]_ser_rx	opt_user_reset	IN
37	mprj_io[6]_ser_tx	—	OUT
38	vssa1	GND	GND
39	vssd1	GND	GND
40	vdda1	VDD	PWR
41	mprj_io[7]	n/c	—
42	mprj_io[8]	hsync_n	OUT
43	mprj_io[9]	vsync_n	OUT
44	mprj_io[10]	rgb[0]	OUT
45	mprj_io[11]	rgb[1]	OUT
46	mprj_io[12]	rgb[2]	OUT
47	vdda1	VDD	PWR
48	mprj_io[13]	rgb[3]	OUT
49	vccd1	VDD	PWR
50	mprj_io[14]	rgb[4]	OUT
51	mprj_io[15]	rgb[5]	OUT
52	vssa1	GND	GND
53	mprj_io[16]	tex_cs_n	OUT
54	mprj_io[17]	tex_sclk	OUT
55	mprj_io[18]	tex_io0	I/O
56	vssio	GND	GND
57	mprj_io[19]	tex_io1	IN
58	mprj_io[20]	tex_io2	IN
59	mprj_io[21]	tex_io3	IN
60	mprj_io[22]	vec_cs_n	IN
61	mprj_io[23]	vec_sclk	IN
62	mprj_io[24]	vec_mosi	IN
63	vccd2	VDD	PWR
64	vddio	VDD	PWR
65	ThermalPad	GND	GND

Power · GND · Raybox input · Raybox output · Raybox bidir · Shared system clock · Unused mprj_io · Caravel-only function · mprj_io / Caravel function

† **Pin 21 / RESETb: always resets Caravel; it directly resets the muxed user designs only when wb_rst_i propagation is enabled. Otherwise, restarted management firmware may reset them; mux/LA configuration selects the behaviour.**

* **Pin 36 / mprj_io[5] — opt_user_reset: optional active-high reset of all muxed user designs when LA[48]=0 and GPIO5 is a user input. Supplied Raybox firmware disables it; Caravel SER_RX shares the pad.**

U1/footprint: QFN-64-1EP_9×9mm_P0.5mm_EP7.65×7.65mm. Pin 65 is the exposed thermal pad, tied to GND on the reference board.

Physical pin order from efabless/caravel_board · Project mapping from algofoogle-multi-caravel (gf180)